

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

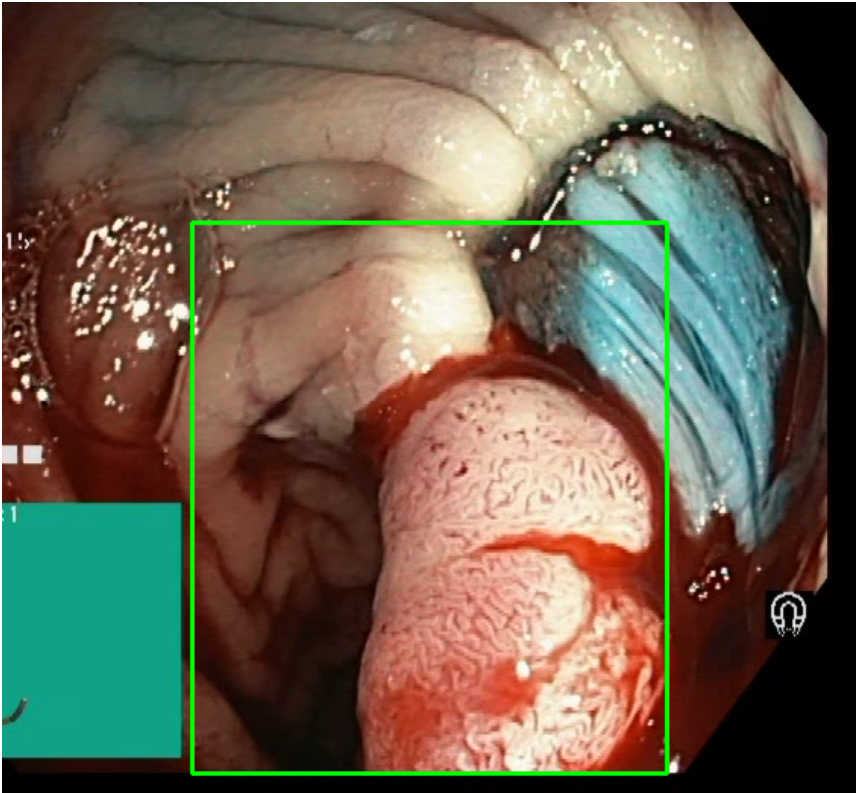
Video ID:	27dd21a7-a8b1-4220-855a-ce397ceb6261
Total Frames:	15013
Frame Rate:	25.0 fps (assumed)
Duration:	600.5 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	3879	✓
RT-DETR Detections	6791	✓
MedSAM2 Detections	6575	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	Count	Percentage
HIGH RISK	1	33.3%
MEDIUM RISK	2	66.7%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 2159 frames)



Feature	Value	Interpretation
Frame Index	8402	Frame 8402 of 15013
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	SERRATED_POLYP	Type confidence: 82.9%
Redness Score	0.146	Color-based vascularization
Relative Radius	46.2%	Polyp size as % of frame diagonal
Texture Score	0.295	Surface morphology
Vessel Visibility	0.032	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket

Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	85.4%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 134 frames)



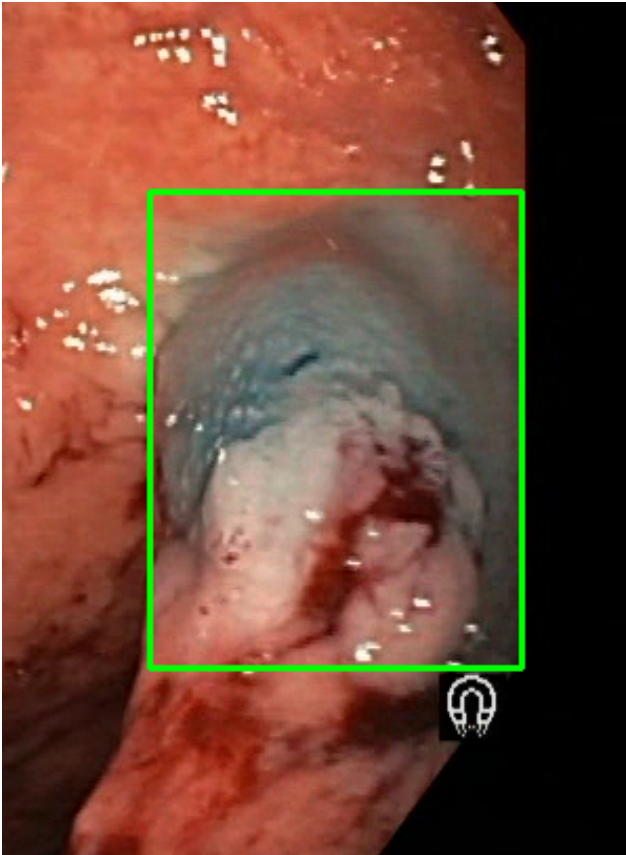
Feature	Value	Interpretation
Frame Index	12390	Frame 12390 of 15013
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.266	Color-based vascularization
Relative Radius	59.2%	Polyp size as % of frame diagonal
Texture Score	0.470	Surface morphology
Vessel Visibility	0.705	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	MALIGNANT_P OLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	85.1%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #3 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 42 frames)



Feature	Value	Interpretation
Frame Index	4723	Frame 4723 of 15013
Detection Method	Detector Consensus (YOLO / RT-DETR / MedSAM2)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 78.3%

Redness Score	0.143	Color-based vascularization
Relative Radius	27.2%	Polyp size as % of frame diagonal
Texture Score	0.220	Surface morphology
Vessel Visibility	0.090	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	83.0%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 1 (33.3%)
- MEDIUM RISK polyps: 2 (66.7%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.